BRIAN · MAIN 48/50 96%

ABERDARE TECHNICAL COLLEGE
DEPARTMENT: INFORMATION COMMUNICATION TECHNOLOGY (ICT)
UNIT NAME: PERFORM PROGRAMMING PRINCIPLES
UNIT CODE: 0613 451 05A
ASSESSMENT: CONTINUOUS ASSESSMENT TEST (CAT 2)
TIME: 1 HOUR 30 MINUTES
TOTAL MARKS: 50

INSTRUCTIONS TO THE CANDIDATE

i. Answer ALL questions.
ii. The marks for each question are shown in brackets.
iii. Write all programs neatly in C and include comments where necessary.
iv. Mobile phones and notes are not allowed in the examination room.

QUESTIONS

1. Define the term *control structure* and name the three main types of control structures in C. (3 marks)

2. Name three input functions in C and state what each function reads. (3 marks)

3. State four criteria used when selecting a data type based on system requirements. (4 marks)

4. Differentiate between a function declaration (prototype) and a function definition. (3 marks)

5. Study the array declaration below and answer the questions that follow. (3 marks)

```
int marks[5] = {50, 60, 70, 80, 90};
```

(a) State the values stored in marks[2] and marks[4]. (2 marks)
(b) State the first and last valid indexes of the array. (1 mark)

6. A three-dimensional array is declared as shown below. (4 marks)

```
int a[2][3][4];
```

(a) Explain the meaning of each of the three indexes. (3 marks)
(b) Calculate the total number of elements the array can store. (1 mark)

7. Study the two-dimensional array below and answer the questions that follow. (5 marks)

```
int scores[3][4] = {
    {12, 8, 15, 20},
    {7, 18, 9, 11},
    {14, 6, 19, 5}
};
```

(a) State the number of rows and columns in the array. (2 marks)
(b) Write the values stored in scores[1][1] and scores[2][3]. (2 marks)
(c) Write a C statement that changes the element in row 2, column 0 to 30. (1 mark)

8. Study the two programs below and answer the questions that follow. (5 marks)

Program 1

```
#include <stdio.h>

void change(int x)
{
   x = 50;
}

int main()
{
   int number = 10;
   change(number);
   printf("Number = %d", number);
   return 0;
}
```

Program 2

```
#include <stdio.h>

void change(int *x)
{
   *x = 50;
}

int main()
{
   int number = 10;
   change(&number);
   printf("Number = %d", number);
   return 0;
}
```

(a) State the output of Program 1 and Program 2. (2 marks)

(b) Name the method of passing arguments used in each program. (2 marks)

(c) Explain why the outputs of the two programs are different. (1 mark)

9. Select the most suitable C data type for storing each of the following data and give one reason to justify each choice. (5 marks)

(a) A student's age.

(b) The population of a country exceeding 2.1 billion.

(c) A percentage score from 0 to 100 on a device with very limited memory.

(d) A measurement requiring approximately 15 digits of precision in a scientific calculation.

(e) A single grade letter such as 'A'.

10. Write a complete C program that accepts a student's marks from the keyboard and uses an if...else if structure to display the grade according to the following criteria. (7 marks)

- 80 and above: Grade A
- 60 to 79: Grade B
- 50 to 59: Grade C
- Below 50: Fail

11. Write a complete C program that performs the following tasks. (8 marks)
(a) Defines a user-defined function called multiply that accepts two integers and returns their product. (3 marks)
(b) Includes a function prototype for the function. (1 mark)
(c) In the main() function, accepts two integers from the user using scanf(), calls the multiply() function, and displays the result. (4 marks)

① A Control Structure is Programming Construct that determines the order in which instructions are executed in a Program.

- Sequence
- Selection
- Iteration

② Scanf() -> Reads formatted inputs such as integers, floating point numbers and characters from the keyboard.

- Getchar() -> Reads a single character from standard input.

③ fgets() -> Reads a line of text, including spaces from Standard input or another input stream.

(3) Range of values
Precision
Memory Requirements
Nature of data.

(1) function declaration
- Specifies the function's name, return type and parameter values

Does not Contain the function Body.

E.g int add(int, int);

function definition
Contains the actual code executed when the function is called

Contains the function Body Enclosed in braces {}

E.g int add

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5]

a Marks [2] = 70 ✓ 1
Marks [4] = 90 ✓ 1
→ indexing in C starts in at index 0

b] First valid index: 0 ✓ 1
Last valid index: 4 ✓ 1

6]

① First index [2]: Represents 2 blocks or group
② Second index [3]: Represents 3 block
③ Third index [4]: Represents 4 Columns within each Row.

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b] Total elements = Number of blocks x Rows x Colum
= 2 x 3 x 4
= Total = 24 ✓ 1

7

a] Number of Rows: 3 ✓ 2
Number of Columns: 4

b] Scores [1][1] = 15 ✓
Scores [2][3] = 5 ✓ 2
Explanation: C arrays start at 0.

c] Scores [2][0] = 30 ✓ 1

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Program 1 Output

Number =

Program 2

b] Program 1 -> Call By value.
Program 2 : Passing an address using a pointer.

6

in Program 1, a Copy of number is passed to the function, so changing x does not change the original

In Program 2, the address of number is passed to the function using *x = so changes the Original variable.

Q] a] A student's age
Answer : int
Reason : Age is normally stored as a whole number

b] Population of a Country Exceeding 2.1 Billion.
Answer : Long Long int
Reason - Can Store very whole large whole numbers, provided the value is within its Supported range

c] 0 - 100
Answer : Unsigned Char
Reason : It Can Store whole numbers from 0 to 100 while using little Memory on typical Systems.

] Answer : double
Reason : It typically provides about 15-16 Significant decimal digits of Precision

9] Answer : char
Reason : It stores a single character

10]
```
#include <stdio.h>

int main()
{
    int Marks;
    printf("Enter Students Marks : ");
    scanf("%d", &Marks);

    if Marks < 0 || Marks > 100)
    {  Printf("invalid Marks");
    }
    Else if (Marks >= 80)
    {  Print("Grade A");
    else if (Marks >= 60)
    {  Printf("Grade B");
    }
    else if (Marks >= 50)
    {  Print("Grade C");
    }
    Else
    {
    Printf("fail");
    }
    Return 0;
    }
```

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11]

```
#include <stdio.h>

int Multiply(int a, int b);

int main()
{
    int num1, num2, result;
    Printf("Enter the first integer: ");
    Scanf("%d", &num1);

    Printf("Enter the second integer: ");
    Scanf("%d", &num2);

    result = Multiply(num1, num2);

    Print f("Product = %d\n", results);
    return 0;
}
int Multiply(int a, int b){
    return a*b;
}
```